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Relevance of Quantum Trace Estimation and Frobenius Polynomial Reconstruction to Physics

A formal statement in response to a question from the Quantum Physics Group

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Abstract

A member of the Quantum Physics Group asked how our paper *Quantum Trace Estimation and Frobenius Polynomial Reconstruction* [CP] is relevant to physics. The paper asserts no physical model, and this note does not add one. Its relevance consists of six precise mathematical statements. Three belong to quantum information physics and are proved or cited theorems: the estimator is a one-clean-qubit (DQC1) measurement (Statement 1); its cost has Heisenberg scaling, $\Theta(1/\delta)$ coherent uses against $\Theta(1/\delta^2)$ independent repetitions (Statement 2); and reconstructing a characteristic polynomial from noisy spectral moments is ill-conditioned, with an explicit two-point lower bound (Statement 3). Three are exact identities that coincide with structures used in physics, and we present them as analogies only: the normalised Frobenius operator is a finite-field Hilbert–Pólya operator whose spectrum gives the zeros of the curve's zeta function (Statement 4); its traces have the moments of the compact symplectic ensemble (Statement 5); and the determinant identities used in the paper are free bosonic and fermionic partition-function identities (Statement 6). A Python script checks every computable claim; all checks pass. For validation instructions, see Section 9, Reproducibility.

1 The question and the scope of the answer

The paper [CP] studies the recovery of the Frobenius polynomial $L_C(T)$ of a curve over a finite field from estimates of the power traces $Tr(U^j)$ of a unitary, obtained by a Hadamard test with amplitude estimation. It is a paper in quantum algorithms and computational number theory. Section 7 of [CP] states explicitly that "no thermodynamic Hamiltonian or experimental physics model is asserted".

The question is therefore not whether the paper models a physical system — it does not — but which of its results bear on physics, and in what sense. Table 1 answers this. Statements 1–3 are statements *in* physics, in the sense of quantum information: they concern a physical measurement, its metrological cost and the conditioning of spectral reconstruction. Statements

4–6 are exact mathematical identities that have the *same form* as structures used in physics; they are offered as analogies, and no physical claim rests on them.

	Content	Status	Kind	Check
1	The estimator is a one-clean-qubit (DQC1) trace measurement	Proved [CP, Lemma 3.1]	Quantum information	A
2	Heisenberg scaling $\Theta(1/\delta)$ versus standard quantum limit $\Theta(1/\delta^2)$	Upper bound proved; optimality cited (black box)	Quantum metrology	B
3	Spectral reconstruction from noisy moments is ill-conditioned	Proved [CP, Thm 3.2, 3.4]	Inverse spectral problem	C
4	Frobenius is a finite-field Hilbert–Pólya operator	Theorem (Weil); formulated here	Analogy	D
5	Traces have compact symplectic random-matrix moments	Cited [KS, DS94]; verified numerically	Analogy	E
6	Determinant identities are free-field partition functions	Exact identities	Analogy	F

Table 1. The six statements, their status, and the check in the validation script that tests each.

Notation

C/F_Q is a smooth projective curve of genus g, with Frobenius numerator

$$L_C(T) = \prod_{i=1}^{2g}(1-\alpha_i T), \qquad |\alpha_i| = \sqrt{Q}. \tag{1}$$

The normalised Frobenius $U_C = Q^{-1/2}F$ acts on $H^1 \otimes C$ with eigenvalues $\beta_i = \alpha_i/\sqrt{Q} = e^{i\theta_i}$, and $p_j = Tr(U_C^j)$. For a general $U \in U(d)$ we write $C_U(T) = det(I - TU) = \sum_k c_k T^k$ and $p_j = Tr(U^j)$.

2 Statement 1: a one-clean-qubit measurement

Statement 1 (DQC1 trace measurement). *For $X \in U(n)$, the Hadamard test with a control qubit and the register in the maximally entangled state $|\Phi_n\rangle$ — equivalently, with the register in the maximally mixed state $\rho = I/n$ — accepts with probability*

$$Pr(0) = \frac{1}{2} + \frac{Re\, Tr(X)}{2n}. \tag{2}$$

Proof. $|\Phi_n\rangle$ purifies I/n, and $\langle\Phi_n|(I \otimes X)|\Phi_n\rangle = Tr(X)/n$; expanding the norm of the control-zero branch gives the formula [CP, Lemma 3.1].

Physics. This is the trace-estimation primitive of the DQC1 model of Knill and Laflamme [KL98], in which a single pure qubit interacts with a maximally mixed register. Estimating a normalised trace to additive precision $1/poly$ is complete for that model [SJ08]. The Frobenius setting is the reverse of the regime usually studied: the dimension is small ($n \le 4g$), while the precision required to recover L_C exactly is exponentially small, $\delta_{cong} \asymp gQ^{-g/2}$ [RT, WP].

Validation. Check A.

– For 100 Haar-random $X \in U(n)$, $n = 2, \ldots, 16$, a statevector simulation reproduces the formula to within 2.2×10^{-16}.

3 Statement 2: Heisenberg scaling

Statement 2 (Metrological cost). *Amplitude estimation recovers $Tr(X^j)$ to additive error δ with probability at least $1-\eta$ using*

$$N_{coh} = \Theta(n\delta^{-1} \log(1/\eta)) \tag{3}$$

coherent applications of controlled-X [BHMT; CP, Thm 4.1]. Estimating the same acceptance probability from independent repetitions of the test requires $N_{ind} = \Theta(n^2\delta^{-2} \log(1/\eta))$. In black-box query models the rate $1/\delta$ is optimal [NW, Mo15]; we do not claim optimality for structured unitaries.

Proof. The trace is $2n(a-1/2)$ up to a known shift, so precision δ on the trace is precision $\varepsilon = \delta/(2n)$ on the acceptance probability a. Amplitude estimation with M register states has error at most $2\pi\sqrt{a(1-a)}/M + \pi^2/M^2$ with probability at least $8/\pi^2$ [BHMT, Thm 12], so $M = \Theta(1/\varepsilon)$ suffices; the median of $O(\log(1/\eta))$ runs gives confidence $1-\eta$. Independent repetitions estimate a by a sample mean with standard deviation $\sqrt{a(1-a)/N}$, so $N = \Theta(1/\varepsilon^2)$ samples are needed when a is bounded away from 0 and 1.

Physics. This is the separation between the Heisenberg limit ($N \propto 1/\delta$) and the standard quantum limit ($N \propto 1/\delta^2$) in quantum metrology [GLM06]: coherent use of the unitary converts the trace into a phase that can be read out at the Heisenberg rate.

Precision ε on a	Amplitude estimation (queries)	Independent repetitions (samples)
2^{-3}	63	24
2^{-5}	255	416
2^{-7}	1,023	7,040
2^{-9}	4,095	112,640
2^{-10}	8,191	450,560

Table 2. Queries needed to guarantee precision ε with probability at least $8/\pi^2$ for every amplitude in [0.05, 0.95] (amplitude estimation, exact output law) and samples needed at a = 1/2 (independent repetitions, exact binomial law). Check B.

Validation. Check B.

– Over $\varepsilon = 2^{-3}, \dots, 2^{-10}$, the fitted log–log slopes of required queries against $1/\varepsilon$ are 1.003 for amplitude estimation and 2.031 for independent repetitions. These finite computations are consistent with the two rates; they do not prove them.

4 Statement 3: ill-conditioning of spectral reconstruction

Newton's identities $kc_k = -\sum_{j=1}^{k} p_j c_{k-j}$ define the map Φ: $(p_1, \dots, p_d) \mapsto (c_1, \dots, c_d)$ from spectral moments to characteristic-polynomial coefficients.

Statement 3 (Conditioning of Φ). *(i) Upper bound [CP, Thm 3.2]. If $|\hat{p}_j - p_j| \le \delta$ for all $j \le d$, then*

$$\max_k |\hat{c}_k - c_k| \le 2^d(e^{\delta H_d} - 1), \qquad H_d = \sum_{j=1}^{d} \frac{1}{j}. \tag{4}$$

(ii) Two-point lower bound [CP, Thm 3.4]. Let P be a polynomial, $U_0 = I_d$, $U_\theta = e^{i\theta} I_d$ and $\theta = 1/(d^2 P(d))$. Then $\sup_{j \le d} | Tr\, U_\theta^j - Tr\, U_0^j | \le 1/P(d)$, while for $k = \lfloor d/2 \rfloor$

$$| c_k(U_\theta) - c_k(U_0) | = C(d,k) \, | e^{ik\theta} - 1 | \ge \frac{2}{\pi} k\theta C(d,k), \tag{5}$$

which grows faster than any polynomial in d. Consequently the minimax risk of any estimator of (c_k) from traces known to within $1/P(d)$ is at least $1/3$ for all large d.

Proof. (i) is proved in [CP] by writing $\hat{C} = C_U e^E$ in $C[T]/(T^{d+1})$ with $\|E\|_1 \le \delta H_d$ and $\|C_U\|_1 \le 2^d$. For (ii), $| Tr\, U_\theta^j - Tr\, U_0^j | = d \, | e^{ij\theta} - 1 | \le dj\theta \le 1/P(d)$ for $j \le d$, and $c_k(U) = (-1)^k C(d,k) \lambda^k$ when $U = \lambda I$, so the coefficient gap is $C(d,k) \, | e^{ik\theta} - 1 | \ge C(d,k)(2/\pi)k\theta$ because $| e^{ix} - 1 | \ge 2x/\pi$ for $0 \le x \le \pi$. The midpoint of the two trace vectors is within $1/(2P(d))$ of both, so any estimator errs by at least half the gap on one of the two unitaries — a two-point argument of Le Cam type [LC86].

Physics. The same obstruction governs every reconstruction of a spectrum or density of states from noisy moments: moment and kernel-polynomial methods [WWAF06], and the classical post-processing of phase-estimation data. Statement 3 gives an explicit, dimension-dependent instance in which polynomially accurate moments determine nothing about the middle coefficients.

d	Trace distance	Bound $1/P(d)$	Coefficient gap	Lower bound
8	0.125	0.125	0.547	0.348
16	0.0625	0.0625	25.1	16.0
32	0.0312	0.0312	2.93×10^5	1.87×10^5
64	0.0156	0.0156	2.24×10^{14}	1.42×10^{14}
128	0.0078	0.0078	7.31×10^{32}	4.65×10^{32}

Table 3. Statement 3(ii) with P(d) = d, evaluated in 60-digit arithmetic (check C). The gap already exceeds 1/3 at d = 8.

5 Statement 4: a finite-field Hilbert–Pólya operator

Define the self-adjoint operator H_C on $H^1 \otimes C$, with the Hermitian structure of [FO, Lemma 1], by

$$U_C = e^{i(\log Q) H_C}, \qquad spec(H_C) \subset [-\pi/\log Q, \pi/\log Q]. \tag{6}$$

Statement 4 (Zeros as spectrum). *Let $\zeta_C(s) = Z_C(Q^{-s})$. Then*

$$\zeta_C(s) = 0 \iff s = \frac{1}{2} + i\gamma, \qquad \gamma \in spec(H_C) + \frac{2\pi}{\log Q} Z. \tag{7}$$

In particular, the Riemann hypothesis for C ($| \alpha_i | = \sqrt{Q}$) is equivalent to the unitarity of U_C, that is, to the self-adjointness of H_C.

Proof. The zeros of ζ_C are the zeros of $L_C(Q^{-s})$, since $(1 - Q^{-s})(1 - Q^{1-s})$ contributes poles. $L_C(Q^{-s}) = 0$ if and only if $Q^s = \alpha_i$ for some i. Writing $\alpha_i = \sqrt{Q} e^{i\theta_i}$ gives $s = \frac{1}{2} + i(\theta_i + 2\pi k)/\log Q$, $k \in Z$, and $\theta_i/\log Q$ is an eigenvalue of H_C. If some $| \alpha_i | \ne \sqrt{Q}$, the corresponding zero has real part $\log | \alpha_i | / \log Q \ne 1/2$ and U_C has an eigenvalue off the unit circle, so it is not unitary; the converse is the same computation read backwards. Weil's theorem [W48] gives $| \alpha_i | = \sqrt{Q}$.

Physics. The Hilbert–Pólya programme seeks a self-adjoint operator whose eigenvalues are the imaginary parts of the zeros of the Riemann zeta function; its physical motivation includes Montgomery's pair correlation and the Berry–Keating proposal [Mon73, BK99]. For curves over finite fields the corresponding operator exists and is H_C: here the analogue of the Riemann hypothesis is a theorem, and the power traces $p_j = Tr(U_C^j)$ estimated in [CP] are the spectral traces of $e^{i(\log Q)H_C}$. This is an exact analogy, not a statement about the Riemann zeta function.

Validation. Check D.

– For 13 curves of genus 1–3, $L_C(Q^{-s})$ vanishes at every predicted $s = \frac{1}{2} + i(\theta_i + 2\pi k)/\log Q$, $k \in \{-1,0,1\}$, to relative precision 1.6×10^{-15}, and every zero of L_C gives real part of s equal to $1/2$ to within 10^{-6} (multiple roots limit the numerical accuracy).

6 Statement 5: symplectic random-matrix statistics

By [FO, Lemma 1], U_C is conjugate to an element of the compact symplectic group $USp(2g)$. Put $\eta_j = 1$ if j is even and $j \le 2g$, and $\eta_j = 0$ otherwise.

Statement 5 (Moments of the symplectic ensemble). *(i) Haar moments [DS94]. For Haar-distributed $U \in USp(2g)$,*

$$E[Tr\, U^j] = -\eta_j \quad (j \ge 1), \qquad E[(Tr\, U^j)^2] = j + \eta_j \quad (1 \le j \le g). \tag{8}$$

(ii) Equidistribution [KS]. For natural families of curves of genus g (for example hyperelliptic curves), the conjugacy classes of U_C become equidistributed with respect to Haar measure on $USp(2g)$ as $Q \to \infty$. Consequently, averaged over the family, $\tau_j(C) = Tr(U_C^j)$ has the moments in (i) in the limit $Q \to \infty$.

Proof. (i) is the Diaconis–Shahshahani computation for the symplectic group [DS94], obtained from the Weyl integration formula with density proportional to $\prod_{i<k}(\cos\theta_i - \cos\theta_k)^2 \prod_i \sin^2\theta_i$ over $0 \le \theta_1, \dots, \theta_g \le \pi$. (ii) is the theorem of Katz and Sarnak [KS]; we cite it and do not use it in any proof.

Physics. $USp(2g)$ is the compact ensemble of symplectic symmetry in Dyson's classification [Dy62], used for spectral statistics in quantum chaos [Haa10]. The power traces τ_j are the variables of the spectral form factor $K(j) = E\,|\,Tr\, U^j\,|^2$, and (i) gives its values for $j \le g$. This is again an analogy of form: the random-matrix statistics describe families of curves, not a physical system.

p	Curves	Mean τ_1	Mean τ_2	Mean τ_1^2
101	10,100	below 10^{-17}	−1.000098	0.999902
211	44,310	below 10^{-17}	−1.000022	0.999978
401	160,400	below 10^{-17}	−1.000006	0.999994
Haar $USp(2)$		0	−1	1

Table 4. Averages over all elliptic curves $y^2 = x^3 + ax + b$ over F_p (genus 1, USp(2) = SU(2)), computed by exhaustive point counting (check E).

Validation. Check E.

– (i): Weyl-integration quadrature on $USp(2g)$ for $g = 1, \ldots, 4$ confirms $E[Tr\ U^j] = -\eta_j$ for all $j \leq 2g + 2$ and $E[(Tr\ U^j)^2] = j + \eta_j$ for $1 \leq j \leq g$, to within 10^{-8}. Beyond $j = g$ the second formula fails (for example $g = 1, j = 2$ gives 2, not 3), which is why it is stated only for $j \leq g$.
– (ii): Table 4 shows the genus-1 averages over every elliptic curve in the family approaching the Haar values as p grows. This illustrates, and does not prove, the equidistribution theorem.

7 Statement 6: free-field partition functions

Statement 6 (Determinant identities). *For any square matrix A with eigenvalues λ_ℓ and power sums $s_m = Tr(A^m)$, as formal power series in z,*

$$det(I - zA)^{-1} = \prod_\ell (1 - z\lambda_\ell)^{-1} = exp\left(\sum_{m\geq 1} \frac{s_m z^m}{m}\right) = \sum_{k\geq 0} h_k(\lambda) z^k, \quad (9)$$

$$det(I - zA) = \prod_\ell (1 - z\lambda_\ell) = \sum_{k\geq 0} (-1)^k e_k(\lambda) z^k, \quad (10)$$

where h_k and e_k are the complete and elementary symmetric polynomials. For the normalised Frobenius spectrum, $c_k = b_k Q^{-k/2} = (-1)^k e_k(\beta)$, and Newton's identities convert between the two expansions.

Proof. $log\,(1 - z\lambda)^{-1} = \sum_m \lambda^m z^m/m$; summing over ℓ and exponentiating gives the first identity, and expanding the product gives h_k and e_k. The relations $kh_k = \sum_{m=1}^k s_m h_{k-m}$ and $ke_k = \sum_{m=1}^k (-1)^{m-1} s_m e_{k-m}$ are Newton's identities.

Physics. The first product is the grand-canonical partition function of a system of free bosons with single-mode weights λ_ℓ, and $log\ Z = \sum\ \ s_m z^m/m$ is its cluster (cumulant) expansion; the second is the fermionic counterpart, and $L_C(T)$ is fermionic in this sense. For the Lubotzky–Phillips–Sarnak graph of [CP, §7], A is the non-trivial transition operator and the identity is of the Ihara type studied for quantum graphs [Ter11, KoS97]. These are identities of form; [CP] asserts no Hamiltonian.

Validation. Check F.
– For 12 random integer matrices the identities hold exactly in rational arithmetic (coefficients of $det(I - zA)$ and of its inverse series through degree 8). For 13 curves, $c_k = (-1)^k e_k(\beta)$ holds to relative precision 4.4×10^{-16}.

8 Summary

The relevance of [CP] to physics is precise and limited:

- **Quantum information (Statements 1–3).** The estimator is a DQC1 trace measurement; its cost has Heisenberg scaling; and reconstructing a characteristic polynomial from noisy spectral moments is provably ill-conditioned. These are statements about a physical measurement procedure and its limits.
- **Exact analogies (Statements 4–6).** The normalised Frobenius operator realises, for curves over finite fields, the unitary spectral interpretation of zeta zeros sought by the Hilbert–Pólya programme; its traces carry the moments of the compact symplectic ensemble; and the paper's determinant identities are free-field partition-function identities. No physical claim rests on these.

- **What would make the connection physical.** All of the above presupposes a coherent circuit for U_C ([CP, Assumption 1.3]). Whether such a circuit can be built without first computing $L_C(T)$ is open; for oracles with few ancillas, building it classically is equivalent to computing $L_C(T)$ [FO, Prop. 5, Conj. 1].

9 Reproducibility

All computable claims in this paper are checked by the validation script `physics_relevance_checks.py` [GH], published under the MIT License at:

https://github.com/teknoquants/Relevance-of-Quantum-Trace-Estimation-and-Frobenius-Polynomial-Reconstruction-to-Physics

The script runs six checks, one per statement:

- **Check A (Statement 1):** the Hadamard-test acceptance probability, ½ + Re Tr(X)/(2n), for 100 Haar-random unitaries with n = 2,…,16.
- **Check B (Statement 2):** the queries needed for precision ε, comparing amplitude estimation (exact output law) with independent repetitions (exact binomial law), and the fitted log–log slopes.
- **Check C (Statement 3):** the two-point construction $U_0 = I$, $U_\theta = e^{i\theta}I$ in 60-digit arithmetic, for d = 8,…,128.
- **Check D (Statement 4):** the zeros of $\zeta_C(s)$ at $s = ½ + i(\theta + 2\pi k)/\log Q$, for 13 curves of genus 1–3.
- **Check E (Statement 5):** the Weyl-integration moments of USp(2g) for g = 1,…,4, and averages of τ_1, τ_2 and τ_1^2 over all elliptic curves $y^2 = x^3 + ax + b$ over F_p, for p = 101, 211 and 401.
- **Check F (Statement 6):** exact rational verification of the determinant identities for random integer matrices, and of $c_k = (-1)^k e_k(\beta)$ for 13 curves.

The script uses exact rational and 60-digit arithmetic wherever a statement is exact. Curve averages come from exhaustive point counting, and random-matrix moments from Gauss–Legendre quadrature of the Weyl integration formula. It shares `frobenius_core.py` with the validation package of [FO].

The repository contains:

- the script and core module;
- pinned requirements (numpy 2.5.3, scipy 1.18.1, sympy 1.14.0, mpmath 1.3.0);
- the environment record of the reference run (CPython 3.13.16, Linux x86_64);
- the reference output and console log;
- a SHA256SUMS file covering every file (verify with sha256sum -c SHA256SUMS).

To reproduce the results, run:

```
python3 -m pip install -r requirements.txt
python3 -I physics_relevance_checks.py
```

The run takes about 10 seconds and writes `results/physics_checks.json`. All checks pass in the reference run.

The checks test the statements on finite instances. They do not prove the asymptotic rates in Statement 2 or the equidistribution theorem in Statement 5, which is cited from [KS].

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